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EQUATIONS OVER SOLVABLE GROUPS

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Not any nonsingular equation over a metabelian group has solution in a larger metabelian group. However, any nonsingular equation over a solvable group with a subnormal series with abelian torsion-free quotients has a solution in a larger group with a similar subnormal series of the same length (and an analogous fact is valid for systems of equations).

0. Introduction

A system of equations $\{w_i = 1 \mid i \in I\}$ with coefficients from a group G, where w_i are words in the alphabet $G \sqcup X^{\pm 1}$, and X is a set (the *set of unknowns*), is called *solvable over* G, if there exist a group $\widetilde{G}$ containing G as a subgroup and a retraction of the free product $\widetilde{G} * F(X)$ onto $\widetilde{G}$ containing all elements w_i in its kernel (henceforth, $F(X)$ is the free group with basis X). If the *solution group* $\widetilde{G}$ can be chosen from a class $\mathcal{K}$, then we say that the system *is solvable in* $\mathcal{K}$.

The study of solvability of equations over groups has a long history: see, e.g., [GR62], [Le62], [Ly80], [How81], [B84], [EH91], [How91], [K93], [KP95], [FeR96], [K97], [K99], [CG00], [EdJu00], [IK00], [Juhá03], [K06], [P08] [BK12], [KL12], [KT17], [Ro17], [BE18], [ABA21], [EH21], [NT22], [KM22], and references therein; see also surveys [Ro12], [NRR79], and book [LS80].

A system of equations (possibly infinite and with, possibly, infinitely many unknowns) over a group is called *nonsingular* if the rows composed of the exponent-sums of unknowns in each equation are linearly independent over $\mathbb{Q}$. If these row are linearly independent over the p-element field $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ for each prime p, then we call the system of equations *unimodular*. In particular, one equation with one unknown is

– nonsingular if the exponent sum of the unknown in this equation is nonzero;

– unimodular if this sum is ± 1.

Unimodular equations behave better than arbitrary nonsingular ones: e.g, in [K93] (see also [FeR96]), it was proven that

any unimodular equation over a torsion-free group is solvable over this group;

it is unknown whether a similar statement is valid for arbitrary nonsingular equations.

For nilpotent groups, everything is simple: Shmel'kin's theorem [Sh67] says (in particular) that

any finite nonsingular system of equations over a nilpotent torsion-free group G has a (unique) solution in a nilpotent group $\widetilde{G} \supseteq G$ of the same nilpotency class; namely, $\widetilde{G}$ is the completion of G (if the system is unimodular, then the torsion-free condition can be dropped, and the unique solution exists in G itself).[*)]

For solvable groups, the situation is more complicated: Section 1 contains examples showing (in particular) that

there exists a unimodular equation with one unknown over a metabelian group (that can be chosen finite or, on the contrary, torsion-free), which is not solvable in any larger metabelian group.

(An example on this subject in [Ro17] is incorrect.)

In 1981, Howie suggested the following generalisation of the well-known Kervaire–Laudenbach conjecture.

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[*)] Note that a theorem of Kuz'min [Ku74] (see also [Ku06]) says that *a finitely generated metabelian group G is residually nilpotent if and only if any unimodular equation with one unknown over G has at most one solution in G* (and the same is true for finitely generated abelian-by-nilpotent groups [Ku78]).

Howie conjecture [How81]**.** *Any nonsingular system of equations over any group is solvable over this group.*

It is unknown whether this is true or false, but there are many partial results on this subject (see references above). We suggest to call a class $\mathcal{K}$ of groups a *Howie class* if any nonsingular system of equations over each group from $\mathcal{K}$ has a solution in a larger group from $\mathcal{K}$.

The quasivariety generated by a Howie class is a Howie class too (see Section 2). Therefore, we are mainly interested in *Howie quasivarieties* (i.e. Howie classes which are quasivarieties). Examples of such quasivarieties are

– the variety of all abelian groups, and also the quasivariety of all abelian torsion-free groups (an easy exercise);
– the quasivariety generated by all finite groups ([GR62] + Malcev theorem on local residual finiteness of linear groups [Ma40]).

Examples from Section 1 mentioned above show that the variety of all metabelian groups is not a Howie class. Nevertheless, we prove the following theorem in Section 3.

Main theorem. *The class consisting of all extensions of groups from a Howie quasivariety by abelian torsion-free groups is a Howie quasivariety too.*

In particular, since abelian groups form a Howie quasivariety, an obvious induction leads us to the following fact on solvable groups:

> *for each positive integer n, the class of groups G admitting a subnormal series $G = G_1 \triangleright \ldots \triangleright G_{n+1} = \{1\}$ whose all quotients G_i/G_{i+1} are abelian, and all, except maybe the last one, are torsion-free is a Howie quasivariety, i.e. any nonsingular system of equations over any group from this class has a solution in a larger group from this class.*

(We can drop the words "except maybe the last one" and obtain another valid fact; indeed, we can start with a simple observation: *the trivial quasivariety $\{\{1\}\}$ is a Howie class*, and then use a similar induction argument.) Note, however, that the minimum length of such a series can be larger than the derived length; see Section 1.

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1. Discouraging examples

Lemma 1. *If elements x and y of a metabelian group are such that y^6 and $u = xx^{-y}x^{y^2}$ belong to the commutator subgroup, then $uu^y u^{-y^3} u^{-y^4} = 1$.*

(Henceforth, we use standard abbreviations: $g^h \stackrel{\text{def}}{=} h^{-1}gh$, $g^{-h} \stackrel{\text{def}}{=} h^{-1}g^{-1}h$, $g^{2023h} \stackrel{\text{def}}{=} h^{-1}g^{2023}h$, …)

Proof. The elements $u = xx^{-y}x^{y^2}$ and x are equal modulo the commutator subgroup (in any group); hence, x lies in the commutator subgroup. Thus,

$$uu^y u^{-y^3} u^{-y^4} = \left(xx^{-y}x^{y^2}\right)\left(xx^{-y}x^{y^2}\right)^{y}\left(xx^{-y}x^{y^2}\right)^{-y^3}\left(xx^{-y}x^{y^2}\right)^{-y^4} = xx^{-y^6} = 1.$$

Here, we use two facts:

– conjugates of x commute (because x lies in the commutator subgroup of a metabelian group);
– $(1 - y + y^2)(1 + y - y^3 - y^4) = 1 - y^6$ (which is an elementary identity).

This completes the proof.

Proposition 1. *There exists a metabelian group G and a unimodular equation $w(x) = 1$ over G having no solutions in any metabelian group $\widetilde{G} \supseteq G$. Moreover, the group G can be chosen*

a) *finite (of order 42)*
b) *or, on the contrary, torsion-free and such that any nonsingular system of equations over G has a solution in a larger three-step solvable group.*

Proof. Take a metabelian group $G \ni a, c$ such that $a^6 = 1$ and c is contained in the commutator subgroup of G. By Lemma 1, the unimodular equation $xx^{-a}x^{a^2} = c$ cannot have solutions in larger metabelian groups

if $cc^a c^{-a^3} c^{-a^4} \neq 1$. A natural example of such a group is the group of triangular invertible 2×2-matrices over the group ring $\mathbb{Z}[y]/\big((y^6-1)\cdot\mathbb{Z}[y]\big)$ of the cyclic group of order six:

$$a = \begin{pmatrix} 1 & 0 \\ 0 & y \end{pmatrix},\ c = \begin{pmatrix} 1 & y-1 \\ 0 & 1 \end{pmatrix} = \left[\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, a \right] \text{ and } cc^a c^{-a^3} c^{-a^4} = \begin{pmatrix} 1 & (y-1)(1+y-y^3-y^4) \\ 0 & 1 \end{pmatrix} \neq 1.$$

(The commutator is understood as $[u,v] \stackrel{\text{def}}{=} u^{-1}v^{-1}uv$.)

To construct a group of order 42, it suffices to make the same construction over the seven-element field $\mathbb{F}_7 = \mathbb{Z}/7\mathbb{Z}$, which is a homomorphic image of the ring $\mathbb{Z}[y]/\big((y^6-1)\cdot\mathbb{Z}[y]\big)$, $y \mapsto 5$ (five is the generator of the multiplicative group of this field):

$$G = \left\{ \begin{pmatrix} 1 & u \\ 0 & v \end{pmatrix} \;\middle|\; u, v \in \mathbb{F}_7,\ v \neq 0 \right\},$$

$$a = \begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix},\ c = \begin{pmatrix} 1 & 5-1 \\ 0 & 1 \end{pmatrix} = \left[\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, a \right] \text{ and } cc^a c^{-a^3} c^{-a^4} = \begin{pmatrix} 1 & (5-1)(1+5-5^3-5^4) \\ 0 & 1 \end{pmatrix} \neq 1.$$

To construct a torsion-free group, we modify the first example:

$$G_1 = \left\{ 2^k \begin{pmatrix} 1 & u \\ 0 & y^k \end{pmatrix} \;\middle|\; u \in \mathbb{Z}[\langle y \rangle_6],\ k \in \mathbb{Z} \right\},$$

$$a = 2\begin{pmatrix} 1 & 0 \\ 0 & y \end{pmatrix},\ c = \begin{pmatrix} 1 & y-1 \\ 0 & 1 \end{pmatrix} = \left[\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, a \right] \text{ and } cc^a c^{-a^3} c^{-a^4} = \begin{pmatrix} 1 & (y-1)(1+y-y^3-y^4) \\ 0 & 1 \end{pmatrix} \neq 1.$$

Clearly, the group G_1 is torsion-free, and a^6 lies in its centre. The central product

$$G = G_1 \underset{a^6=f}{\bigtimes} H, \quad \text{where} \quad H = \langle d, e, f \mid [d,e] = f,\ [d,f] = [e,f] = 1 \rangle,$$

of G_1 and the Heisenberg group $H \simeq \mathbf{UT}_3(\mathbb{Z})$ satisfies conditions of Lemma 1 and is torsion-free (because, in the Heisenberg group, the centre $\langle f \rangle$ is isolated). Moreover, G has a subnormal series of length three

$$G \triangleright G_1 \triangleright \left\{ \begin{pmatrix} 1 & u \\ 0 & 1 \end{pmatrix} \;\middle|\; u \in \mathbb{Z}[\langle y \rangle_6] \right\} \triangleright \{1\}$$

whose quotients, obviously, are free abelian groups of ranks two, one, and six. This completes the proof of Proposition 1, because, by virtue of the main theorem, a group with a length-three subnormal series with abelian torsion-free quotients form a Howie class.

(We note parenthetically that, in the last example, there are no length-two subnormal series with abelian torsion-free quotients, because, by the main theorem, such a series would imply the solvability of the equation $xx^{-a}x^{a^2} = c$ in a larger metabelian group, which is not the case, as we have shown.)

2. Quasivarieties and systems of equations

Recall that a *quasivariety* of groups is the class of all groups satisfying a given (finite or infinite) set of *quasi-identities*, i.e. sentences of the form

$$\forall x, y, \ldots\ v_1(x,y,\ldots) = 1 \,\&\, \ldots \,\&\, v_k(x,y,\ldots) = 1 \Longrightarrow w(x,y,\ldots) = 1,$$

where v_i and w are some elements of the free group $F(x,y,\ldots)$. For example, the quasivariety of abelian torsion-free groups can be defined by the following quasi-identities (the last one is, in fact, an identity): $\forall x\ x^2 = 1 \Longrightarrow x = 1$, $\forall x\ x^3 = 1 \Longrightarrow x = 1$, $\ldots$, $\forall x, y\ 1 = 1 \Longrightarrow [x,y] = 1$. Another (equivalent) definition of the notion of quasivariety can found below, in the beginning of the proof of Proposition 2. For more details about quasivarieties, see books [Bu02] and [Go99]; we recall only that

> *the class consisting of all possible extensions of groups from a quasivariety by groups from another quasivariety is a quasivariety* [Ma67] (called the *product* of two quasivarieties).

Therefore, to prove the main theorem, it suffices to show that the class of extensions (in that theorem) is a Howie class.

Proposition 2. *For any quasivariety $\mathcal{Q}$, the class of groups*

$$\mathbf{H}(\mathcal{Q}) = \left\{ G \in \mathcal{Q} \;\middle|\; \begin{array}{l} \text{any nonsingular system of equations over } G \\ \text{has a solution in a larger group from } \mathcal{Q} \end{array} \right\}$$

coincides with the class

$$\mathbf{H}_{\text{fin}}(\mathcal{Q}) = \left\{ G \in \mathcal{Q} \;\middle|\; \begin{array}{l} \text{any finite nonsingular system of equations over } G \\ \text{has a solution in some larger group from } \mathcal{Q} \end{array} \right\}$$

and is a quasivariety.

Proof. By a theorem of Malcev (see [Ma70], Theorem 5.11.4),

> *quasivarieties of groups are the same as nonempty classes of groups closed with respect to subgroups and reduced (filtered) products.*

The closedness of classes $\mathbf{H}(\mathcal{Q})$ and $\mathbf{H}_{\text{fin}}(\mathcal{Q})$ with respect to these operations is fairly obvious (to obtain a solution group for a system of equations over a reduced product, we can take the reduced product (with respect to the same filter) of solution groups for the corresponding systems of equations over the factors).

The coincidence of the classes $\mathbf{H}(\mathcal{Q})$ and $\mathbf{H}_{\text{fin}}(\mathcal{Q})$ is a special case of the Gödel–Malcev compactness theorem; but for reader's convenience, we give a sketch of a direct proof:

- consider an (infinite) nonsingular system of equations over a group $G \in \mathbf{H}_{\text{fin}}(\mathcal{Q})$ with an (infinite) set of unknowns X;
- for each finite subsystem W, take its solution group $G_W \in \mathcal{Q}$;
- take the Cartesian product of all G_W;
- and choose a set (of values of the unknowns) $\widetilde{X}$ in this product: their coordinates with index W form a solution to the finite system W;
- on the set of indices $\{W\}$, consider the filter F consisting of all sets containing all "sufficiently large indices": $F = \big\{ M \mid \exists W \; M \supseteq \{ W' \mid W' \supseteq W \} \big\}$;
- the group G is embedded diagonally into this reduced product $\widetilde{G} = \left(\bigtimes\limits_{W} G_W \right) / F \in \mathcal{Q}$;
- the set $\widetilde{X} \subset \widetilde{G} \in \mathcal{Q}$ forms a solution to the entire system (because, for each equation $w \in W$, the coordinates $\widetilde{X}_W$ form a solution to w if $W \ni w$; and this set of coordinates $\{W \mid W \ni w\}$ lies in the filter F);
- thus, $G \in \mathbf{H}(\mathcal{Q})$, and this completes the proof of Proposition 2.

Proposition 2 shows, in particular, that

1) the quasivariety generated by a Howie class is a Howie class itself;
   indeed, for the quasuvariety $\mathbf{qvar}\,\mathcal{K}$ generated by a Howie class $\mathcal{K}$, we obtain that $\mathbf{H}(\mathbf{qvar}\,\mathcal{K})$
   - is a quasivariety (by Proposition 2)
   - and contains $\mathcal{K}$ (since $\mathcal{K}$ is a Howie class, i.e. nonsingular systems of equations over groups from $\mathcal{K}$ are solvable in $\mathcal{K} \subseteq \mathbf{qvar}\,\mathcal{K}$);
   - hence, $\mathbf{H}(\mathbf{qvar}\,\mathcal{K}) \supseteq \mathbf{qvar}\,\mathcal{K}$ (as $\mathbf{qvar}\,\mathcal{K}$ is the least quasivariety containing $\mathcal{K}$);
   - the last inclusion is an equality actually, because the inverse inclusion holds by the definition of $\mathbf{H}(\ldots)$

   (therefore, it is natural to study Howie quasivarieties rather than arbitrary Howie classes);
2) in the proof of the main theorem, we can restrict ourselves to finite systems of equations, i.e. the main theorem is equivalent to the following (formally weaker) statement.

Main theorem (a simplified form)**.** *Suppose that a group G contains a normal subgroup B from a Howie quasivariety $\mathcal{K}$, and the quotient group $G/B = A$ is abelian and torsion-free. Then any finite nonsingular system of equations $\{w_i = 1 \mid i \in I\}$ (where $w_i \in G * F(X)$ and I is a finite set) has a solution in some group $\widetilde{G} \supseteq G$, which is also an extension of a group from $\mathcal{K}$ by an abelian torsion-free group.*

3. Proof of the main theorem

Over the group A, we have the induced system of equations: $\{\overline{w}_i = 1 \mid i \in I\}$, where $\overline{w}_i \in A * F(X)$ are images of w_i under the natural homomorphism $G * F(X) \to A * F(X)$.

The first simple observation is that

we can assume that the induced system of equations $\{\overline{w}_i = 1\}$ has a solution in A.

Indeed, this system of equations over $A = G/B$ has solution in some larger torsion-free abelian group $\widetilde{A}$ (because the class of abelian torsion-free groups is a Howie class); therefore, we can embed G into the Cartesian (= unrestricted) wreath product $B \bar{\wr} A$ by the Kaloujnine–Krasner theorem [KK51], and then embed the wreath product $B \bar{\wr} A$ into $B \bar{\wr} \widetilde{A}$; now, we replace A with $\widetilde{A}$, and B with the base of the wreath product (using closedness of the quasivariety $\mathcal{K}$ with respect to Cartesian powers); we obtain what we want. As a byproduct of this argument, we come to the conclusion that

we can assume that the initial extension splits: $G = A \leftthreetimes B$.

The next observation is that

we can assume that a solution to the induced system of equations $\{\overline{w}_i = 1\}$ in A is the set of identity elements.

Indeed, this can be achieved by an obvious change of variables (because we assume already that some solution exists in A).

This observation means that w_i can be rewritten as words in the alphabet $B \sqcup \bigsqcup\limits_{a \in A} X^{\pm a}$:

$$G * F(X) \ni w_i = f(v_i), \quad \text{where } v_i \in B * F\left(\bigsqcup_{a \in A} X_a\right) \text{ are some words,}$$

and the homomorphism $f \colon B * F\left(\bigsqcup\limits_{a \in A} X_a\right) \to G * F(X)$ sends $X_a \ni x_a$ to $x^a \stackrel{\text{def}}{=} a^{-1} x a$ and $B \ni b$ to b.

On the group $B * F\left(\bigsqcup\limits_{a \in A} X_a\right)$, the group A acts naturally (on the right): $b \circ a = a^{-1} b a$ and $x_{a'} \circ a = x_{a'a}$. The key observation (whose proof is postponed, see Lemma 3 below) is that

the system of equations $\{v_i \circ a = 1 \mid i \in I,\ a \in A\}$ over B (with unknowns $\bigsqcup\limits_{a \in A} X_a$) is nonsingular. $(*)$

Now everything is simple.

- Since the system of equations $\{v_i \circ a = 1 \mid i \in I,\ a \in A\}$ over B is nonsingular, it has a solution in a group $\widetilde{B} \supseteq B$ from $\mathcal{K}$ (because $\mathcal{K}$ is a Howie class);
- i.e. the quotient group $U = \left(B * F\left(\bigsqcup\limits_{a \in A} X_a\right)\right) \Big/ \langle\langle \{v_i \circ a = 1 \mid i \in I,\ a \in A\} \rangle\rangle$ admits a homomorphism to $\widetilde{B}$ whose restriction to B is identity.
- Therefore, the quotient group $\widehat{B} \supseteq B$ of U by the intersection of kernels of all homomorphisms to $\widetilde{B}$
  - contains B and a solution to the system of equations $\langle\langle \{v_i \circ a = 1 \mid i \in I,\ a \in A\} \rangle\rangle$ over B (obviously);
  - is contained in $\mathcal{K}$ (because $\widehat{B}$ is sub-Cartesian product of subgroups of $\widetilde{B} \in \mathcal{K}$, and the class $\mathcal{K}$ is a quasivariety.
- It remains to note that the natural action of A on U (extending the action of A on B) induces an action of A on $\widehat{B}$ (because the intersection of kernels of all homomorphisms from a group to another one is a characteristic subgroup).
- Therefore, the corresponding semidirect product $\widetilde{G} = A \leftthreetimes \widehat{B} \supseteq A \leftthreetimes B = G$ is the required group containing a solution to the initial system and being an extension of a group from $\mathcal{K}$ by an abelian torsion-free group.

It remains to prove $(*)$. This fact is a corollary of the following lemma for

- $R = \mathbb{Z}[A]$ is the group ring of A (this ring has no zero divisors, as is well known — we leave a proof to the reader as an easy exercise),
- M is the free abelian group with basis $\bigsqcup\limits_{a\in A} X_a$,
- $\{m_1, \dots, m_k\} \subset M$ is the image of an arbitrary finite subset of $\{v_i \mid i \in I\}$ under the natural homomorphism $B * F\left(\bigsqcup\limits_{a\in A} X_a\right) \to M$ (with the kernel (the commutator subgroup) $\cdot\, B$),
- I is the *fundamental ideal*, i.e. the kernel of the homomorphism $\mathbb{Z}[A] \to \mathbb{Z}$ sending A to 1.

Lemma 2. *Suppose that elements $m_1, \dots, m_k$ of a free module M over an associative commutative ring R without zero divisors are linearly dependent over R. Then, for any prime ideal $I \lhd R$, the elements $m_i + IM$ of the quotient module $M/(IM)$ over the quotient ring R/I are linearly dependent over R/I.*

Proof. Let us choose some basis $e_1, \dots, e_n$ of the module M (which can surely be assumed finitely generated). The linear dependence of elements $m_i = \sum\limits_j r_{ij} e_j$ means that all the minors of order k in the matrix (r_{ij}) are zero (because the ring R embeds into a field). Therefore, all minors of order k in the matrix $(r_{ij} + I)$ are zero. Therefore, $m_i + IM = \sum\limits_j (r_{ij} + I)(e_j + IM)$ are linearly dependent (because the ring R/I embeds into a field too). This completes the proofs of the lemma and theorem.

REFERENCES

[ABA21] M. F. Anwar, M. Bibi, M. S. Akram, On solvability of certain equations of arbitrary length over torsion-free groups, Glasgow Mathematical Journal, 63:3 (2021), 651-659. See also arXiv:1903.06503 .

[BK12] D. V. Baranov, A. A. Klyachko, Economical adjunction of square roots to groups, Siberian Math. Journal, 53:2 (2012), 201-206. See also arXiv:1101.3019 .

[BE18] M. Bibi, M. Edjvet, Solving equations of length seven over torsion-free groups, Journal of Group Theory, 21:1 (2018), 147-164.

[B80] S. D. Brodskii, Equations over groups and group with a single defining relation, Russian Math. Surveys, 35:4 (1980), 165-165.

[B84] S. D. Brodskii, Equations over groups and group with one relator, Siberian Math. Journal, 25:2 (1984), 235-251.

[Bu02] A. I. Budkin, Quasivarieties of groups [in Russian], Altay State Univ., Barnaul, 2002.

[CG00] A. Clifford, R. Z. Goldstein, Equations with torsion-free coefficients, Proc. Edinburgh Math. Soc., 43:2 (2000), 295-307.

[EH91] M. Edjvet, J. Howie, The solution of length four equations over groups, Trans. Amer. Math. Soc., 326:1 (1991), 345-369.

[EH21] M. Edjvet, J. Howie, On singular equations over torsion-free groups, International Journal of Algebra and Computation, 31:3 (2021), 551-580. See also arXiv:2001.07634 .

[EdJu00] M. Edjvet, A. Juhász, Equations of length 4 and one-relator products, Math. Proc. Cambridge Phil. Soc., 129:2 (2000), 217-230.

[FeR96] R. Fenn, C. Rourke, Klyachko's methods and the solution of equations over torsion-free groups, L'Enseignment Mathématique, 42 (1996), 49-74.

[GR62] M. Gerstenhaber, O.S. Rothaus, The solution of sets of equations in groups, Proc. Nat. Acad. Sci. USA, 48:9 (1962), 1531-1533.

[Go99] V.A. Gorbunov, Algebraic theory of quasivarieties. Siberian School of Algebra and Logic. Consultants Bureau, New York, 1998.

[How81] J. Howie, On pairs of 2-complexes and systems of equations over groups, J. Reine Angew Math., 1981:324 (1981), 165-174.

[How91] J. Howie, The quotient of a free product of groups by a single high-powered relator. III: The word problem, Proc. Lond. Math. Soc., 62:3 (1991), 590-606.

[IK00] S. V. Ivanov, A. A. Klyachko, Solving equations of length at most six over torsion-free groups, Journal of Group Theory, 3:3 (2000), 329-337.

[Juhá03] Juhász A. On the solvability of a class of equations over groups, Math. Proc. Cambridge Phil. Soc., 135:2 (2003), 211-217.

[K93] A. A. Klyachko, A funny property of sphere and equations over groups, Communications in Algebra, 21:7 (1993), 2555-2575.

[K97] A. A. Klyachko, Asphericity tests, International Journal of Algebra and Computation, 7:4 (1997), 415-431.

[K99] A. A. Klyachko, Equations over groups, quasivarieties, and a residual property of a free group, Journal of Group Theory, 2:3 (1999), 319-327.

[K06] A. A. Klyachko, How to generalize known results on equations over groups, Math. Notes, 79:3 (2006), 377-386. See also arXiv:math.GR/0406382 .

[KL12] A. A. Klyachko, D. E. Lurye, Relative hyperbolicity and similar properties of one-generator one-relator relative presentations with powered unimodular relator, Journal of Pure and Applied Algebra, 216:3 (2012), 524-534. See also arXiv:1010.4220 .

[KM22] A. A. Klyachko, M. A. Mikheenko, Yet another Freiheitssatz: Mating finite groups with locally indicable ones, Glasgow Mathematical Journal (to appear). See also arXiv:2204.01122 .

[KP95] A. A. Klyachko, M. I. Prishchepov, The descent method for equations over groups, Moscow Univ. Math. Bull. 50:4 (1995), 56-58.

[KT17] A. A. Klyachko, A. B. Thom, New topological methods to solve equations over groups, Algebraic and Geometric Topology, 17:1 (2017), 331-353. See also arXiv:1509.01376 .

[KK51] M. Krasner, L. Kaloujnine, Produit complet des groupes de permutations et le problème d'extension de groupes. III, Acta Sci. Math., 14 (1951), 69-82.

[Ku74] Yu. V. Kuz'min, Residual properties of metabelian groups [in Russian], Algebra i Logika, 13:3 (1974), 300-310.

[Ku78] Yu. V. Kuz'min, Some approximation properties the variety $\mathfrak{AM}_c$, Russian Math. Surveys, 33:4 (1978), 257-258.

[Ku06] Yu. V. Kuz'min, Homological group theory [in Russian]. Moscow: Faktorial, 2006.

[Le62] F. Levin, Solutions of equations over groups, Bull. Amer. Math. Soc., 68:6 (1962), 603-604.

[Ly80] R. C. Lyndon, Equations in groups, Bol. Soc. Bras. Math., 11:1 (1980), 79-102.

[LS80] R. Lyndon, P. Schupp, Combinatorial group theory. Springer, 1977.

[Ma40] A. Malcev, On isomorphic matrix representation of infinite groups [in Russian], Mat. Sb., 8(50):3 (1940), 405-422.

[Ma67] A. I. Maltsev, Multiplication of classes of algebraic systems, Siberian Math. J., 8:2 (1967), 254-267.

[Ma70] A. I. Mal'cev, Algebraic systems. Springer-Verlag, New York, Heidelberg, and Berlin, 1973.

[NT22] M. Nitsche, A. Thom, Universal solvability of group equations, Journal of Group Theory, 25:1 (2022), 1-10. See also arXiv:1811.07737 .

[NRR79] G. A. Noskov, V. N. Remeslennikov, V. A. Roman'kov, Infinite groups, J. Soviet Math., 18:5 (1982), 669-735.

[P08] V. G. Pestov, Hyperlinear and sofic groups: A brief guide, Bull. Symb. Log., 14:4 (2008), 449-480. See also arXiv:0804.3968 .

[Ro12] V. A. Roman'kov, Equations over groups, Groups-Complexity-Cryptology, 4:2 (2012), 191-239.

[Ro17] V. A. Roman'kov, On solvability of regular equations in the variety of metabelian groups, Applied discrete mathematics, 36 (2017), 51-58.

[Sh67] A. L. Shmel'kin, Complete nilpotent groups [in Russian], Algebra i Logika Seminar, 6:2 (1967), 111-114.